Question #1 of 75

Question ID: 1675132

An investor is looking to establish an option strategy that does not involve a position in an underlying share, but would provide a modest potential for gain in case of the share price decline, while limiting losses if the share price rose. Which of the following option strategies would *best* achieve these objectives?

A) Bear call spread. ✓

B) Bull put spread. ✗

C) Zero-cost collar. ✗

Explanation

Spreads do not involve a position in the underlying share. A bear call spread involves buying a higher strike price call and selling a lower strike price call. Because the lower strike price option is more expensive, the maximum gain from this trade is the credit received from the net premiums. In case of a share price decline, the options would become out-of-the-money and the investor keeps the net premium. In case of a share price increase, the investor's loss is limited to the difference between the two call option strike prices, less the initial credit received from the net premiums.

A bull put spread is not appropriate as it benefits from a share price increase, not a decline.

A zero-cost collar is not appropriate because it, in addition to trading a call and a put, involves a position in the underlying share.

(Module 16.6, LOS 16.f)

Question #2 of 75

Question ID: 1675184

An uncovered (short) put option position on an asset can be hedged by:

A) buying a call. ✗

B) short selling the asset. ✓

C) buying the asset. ✗

Explanation

An uncovered short put position would obligate the put seller to buy the asset if the put is exercised. Hedging this position can be accomplished by (short) selling the asset.

(Module 16.12, LOS 16.i)

Question #3 of 75

Question ID: 1675114

Suppose an investor purchases a stock at $41 today and sells (takes a short position in) an $45 call, for a $3 premium. The investor's breakeven price is *closest* to:

A) $44.

B) $37.

C) $38.

Explanation

Because the investor receives the $3 premium from selling the call, any amount of stock price increase will leave the investor with a small net gain. The stock price would have to decline by the premium value for the investor to reach breakeven, which will occur at $38 (at $38, the investor has a $3 loss on the stock, exactly offset by the $3 premium received from selling the call).

(Module 16.3, LOS 16.b)

Question #4 of 75

Question ID: 1675123

Assume that the current price of a stock is $100. A call option on that stock with an exercise price of $97 costs $7. A call option on the stock with the same expiration and an exercise price of $103 costs $3. Using these options, what is the cost of entering into a long bull spread on this stock?

A) $0.

B) $1.

C) $4.

Explanation

The buyer of a bull spread buys the call with an exercise price below the current stock price and sells the call option with an exercise price above the stock price. The cost of the strategy is the difference between the cost of buying the option with the lower exercise price and selling the option with the higher exercise price, which is $7 – $3 = $4 to enter into this strategy.

(Module 16.6, LOS 16.f)

Question #5 of 75

Question ID: 1675185

An investor owns DEF shares that trade at $28 on January 15. The investor prefers not to sell the shares but would like to generate additional income from writing call options on the shares. He expects a maximum price volatility of 10% over the next two months. Which of the following strategies is *most appropriate* for the investor? (Assume that options expire on the 15th of each month.)

A) Sell MAR $31 calls. ✓

B) Sell MAR $28 calls. ✗

C) Sell FEB $31 calls. ✗

Explanation

The investor could use a covered call strategy. Selling the 2-month (MAR) call option will generate a higher premium than the 1-month (FEB) call option, so it would satisfy the investor's desire for generating additional income. Given that the investor does not want to sell his shares, the option strike price should be set outside of the 10% volatility band (the strike price should, therefore, be greater than $28 × 1.1 = $30.80 to ensure the option does not expire in the money).

(Module 16.12, LOS 16.i)

Question #6 of 75

Question ID: 1675163

An analyst noticed that following a period of low volatility, implied volatility has picked up for out-of-the-money (OTM) put options, while the price of OTM call options declined. Which of the following terms *best* describes this scenario?

A) Risk reversal. ✗

B) Volatility smile. ✗

C) Volatility skew. ✓

Explanation

A *volatility skew* describes a scenario where implied volatility increases for OTM puts and decreases for OTM calls. OTM puts are more desirable as insurance against market declines, and the increased desirability increases the price, and therefore volatility, of OTM put options.

A *risk reversal* strategy combines long (short) calls and short (long) puts on the same underlying. A *volatility smile* describes a scenario where further-from-at-the-money options have higher implied volatilities.

(Module 16.11, LOS 16.h)

Question #7 of 75

Question ID: 1675121

Buying a call option on a short position in an underlying security has the same payoff structure at option expiration as a:

A) long put. ✓

B) short call. ✗

C) long put combined with a short call. ✗

Explanation

A short position in a security gains value if the security price declines and loses value if the security price increases. Buying a call on the security hedges the risk of a price increase in the security. This position has the same payoff structure at expiration as a long put, which benefits from a price decline of the underlying security but is not exposed to a price increase of the underlying.

(Module 16.5, LOS 16.e)

Question #8 - 11 of 75

Question ID: 1675152

Kovac is considering a bear put spread strategy. The strategy is *most appropriate* for:

A) Stock 2. ✗

B) Stock 1. ✗

C) Stock 3. ✓

Explanation

A bear put spread is appropriate when a modest stock price decline is expected and there is very limited potential for a price increase. The option strategy involves selling a less expensive, lower strike price put and buying a more expensive, higher strike price put on the same stock with the same expiry to protect against price decreases.

The bear put is a debit spread (net cash outflow) with the cash outflow being equal to the difference between the two option premiums. The maximum gain is when the stock price is at or below the lower put strike price. The maximum profit is the difference between strikes minus the net premium paid.

(Module 16.6, LOS 16.f)

Question #9 - 11 of 75

Question ID: 1675153

Kovac is also considering a short straddle strategy. This strategy is *most appropriate* for:

A) Stock 1. ✗

B) Stock 3. ✗

C) Stock 2. ✓

Explanation

Short straddles involve selling both a call and a put with the same exercise prices and maturity on the same underlying stock. They are neutrality plays because they work well in neutral (flat) markets.

The seller of the options receives two premiums (one from the call and one from the put). The potential loss is very large if the stock price moves either up or down significantly (especially if by more than the premium income earned). The loss from a decline in stock price is limited because the stock price cannot fall below zero. However, the loss from an increase in stock price is unlimited because the stock can, in theory, rise infinitely.

(Module 16.6, LOS 16.f)

Question #10 - 11 of 75

Question ID: 1675154

For this question only, assume that Kovac decides to establish a long calendar spread for Stock 1 using call options. A long calendar spread is:

A) a credit spread, and the maximum loss is limited by the call strike price. ✗

B) a debit spread, and the maximum gain is limited by the call strike price. ✗

C) a debit spread, and the maximum gain is unlimited. ✓

Explanation

A long calendar spread using calls is a debit spread, meaning there is a net cash outlay to establish this strategy. It involves buying a more expensive, longer-dated call option and selling a less expensive, shorter-dated call option with the same strike price and on the same underlying security.

If the shorter-dated call expires out of the money (OTM), the maximum gain on the longer-dated long call is unlimited. The strategy performs best during stable markets or rising volatilities.

The maximum loss on the strategy is the net premium paid.

(Module 16.10, LOS 16.g)

Question #11 - 11 of 75

Question ID: 1675155

Recently, Kovac established a protective put position on a stock other than Stocks 1, 2, and 3. Which of the following statements about a protective put is *most accurate*?

A) The net exposure is equivalent to a long call. ✓

B) The maximum gain is the put strike price. ✗

C) Pre-expiry, the position delta is between −1 and 0. ✗

Explanation

A protective put involves owning the underlying stock and buying a put option on the stock. The position protects against a stock price decline as the put option limits the downside loss but allows an uncapped gain from a price increase. As a result, the net exposure is equivalent to a long call.

The maximum gain on a protective put is unlimited. Pre-expiry, the position delta is between 0 and +1 (similar to a long call).

(Module 16.9, LOS 16.d)

Question #12 of 75

Question ID: 1675112

Which of the following statements regarding covered call options on an underlying stock is *most correct*?

A) The strategy is used to generate additional portfolio income by speculating that the underlying stock price will change significantly. ✗

B) The maximum loss is equal to the purchase price of the stock less the call premium. ✓

C) The breakeven price is equal to the strike price of the stock less the call premium. ✗

Explanation

The maximum loss is the cost of the stock, offset by the premium earned by selling the call. The maximum loss is denoted as $S_0 - C_0$.

The breakeven price is equal to the cost of the stock less the call premium, denoted as $S_0 - C_0$.

(Module 16.3, LOS 16.b)

Question #13 of 75

Question ID: 1675187

An investor is expecting low volatility in the price of a stock that she owns without a clear up or down trend. The *best* course of action for the investor to take would be to:

A) enter into a collar. ✗

B) sell out-of-the-money calls. ✓

C) buy at-the-money puts. ✗

Explanation

Selling calls (a covered call strategy) provides premium income for the investor. The out-of-the-money calls likely won't be exercised if the stock price changes are small.

At-the-money puts are expensive, and given an expectation of only low price volatility, the investor does not need protection against a price decline. For these reasons, a collar (a protective put and a covered call) is also not appropriate.

(Module 16.12, LOS 16.j)

Question #14 - 17 of 75

Question ID: 1675141

Which of the following is *most likely* equivalent to a synthetic long forward position that will offset Mult's short position in the forward contract on Zuhu shares?

A) Long a call option and short a put option on Zuhu shares with an exercise price of EUR 30 and that expire in three months. ✓

B) Long a call option and long a put option on Zuhu shares with an exercise price of EUR 30 and that expire in three months. ✗

C) Buy 80,000 shares of Zuhu at the current market price of EUR 28.5 by borrowing EUR 2,280,000 for three months. ✗

Explanation

To offset the short forward position on Zuhu shares, Mult can create a synthetic long forward position by going long a call option and going short a put option with an exercise price equal to the price of the short forward contract (EUR 30). At expiry, if the stock price is higher than EUR 30, Mult will exercise the call option and buy the shares for EUR 30; if the stock price is lower than EUR 30, the owner of the put will exercise the option, and Mult will have to buy the shares at EUR 30.

While borrowing EUR 2,280,000 (= €28.5 × 80,000) and buying 80,000 shares of Zuhu will hedge Mult's short forward position, such a strategy is not equivalent to a synthetic forward position.

To create a synthetic long forward position, Mult should short the put option.

(Module 16.1, LOS 16.a)

Question #15 - 17 of 75

Question ID: 1675142

Based on **Exhibit 1: Selected Information on Zuhu* Options** and in line with Mult's expectation, the maximum loss and profit on the covered call position that Mult should implement to reap the highest benefit are *closest* to:

A) EUR 30.1 and EUR 1.9, respectively.

B) EUR 31.5 and EUR 0.5, respectively.

C) EUR 29.6 and EUR 2.4, respectively.

Explanation

A covered call strategy is implemented by writing a call option on a share that is owned by the investor. Given that Mult expects the stock price to remain stable for the next two months, to generate the highest profit, Mult should write the out-of-the-money call option that expires in two months—the call option with the exercise price of EUR 32 expiring in March.

The maximum loss and the maximum profit on a covered call are calculated as

$$\text{maximum loss} = S_0 - c_0$$

$$\text{maximum profit} = (X - S_0) + c_0$$

where S_0 is the stock price when the option position was opened, X is the option exercise price, and c_0 is the call premium. Thus, the correct answer follows:

Maximum loss = 31.5 – 1.4 = EUR 30.1.

Maximum profit = (32 – 31.5) + 1.4 = EUR 1.9.

The April call option is erroneously considered instead of the March call option, where the result is the maximum loss = 31.5 – 1.9 = EUR 29.6 and the maximum profit = 32 – 31.5 + 1.9 = EUR 2.4. Because Mult expects Zulu's share price to remain stable for the next two months only, Mult will not choose an option with an expiry longer than two months.

The option premium is excluded from the calculations of maximum loss and maximum profit.

(Module 16.3, LOS 16.b)

Question #16 - 17 of 75

Question ID: 1675143

Based on **Exhibit 2: Selected Information Pertaining to Options on XVOX* (In USD)** and Mult's expectation pertaining to XVOX, the straddle will *most likely* become profitable if XVOX's share price at expiration is:

A) lower than $26.05 or higher than $33.95. ✓

B) between $26.05 and $33.95. ✗

C) lower or higher than $30. ✗

Explanation

Because Mult expects the share price of XVOX to move sharply upward or downward, a straddle strategy is suitable. A straddle strategy consists of buying a call option and a put option with the same exercise price and maturity.

The premiums for the call and put options are $2.15 and $1.80, respectively, resulting in a net cost of $3.95. Therefore, for the straddle to be profitable, it will have to generate a minimum payoff of $3.95.

If the stock price falls below $30, the put option will be exercised, and if the stock price increase above $30, the call option will be exercised. Thus, to generate a profit on the straddle, the stock price would have to be either lower than 30 – 3.95 = $26.05, or higher than 30 + 3.95 = $33.95.

If the share price at expiration is lower or higher than $30, the strike price is not adjusted for the call and put premiums. If the share price at expiration is between $26.05 and $33.95, this represents a price range over which the straddle would generate a loss

(Module 16.6, LOS 16.f)

Question #17 - 17 of 75

Question ID: 1675144

Based on **Exhibit 3: Selected Information Related to the Suxzi AG Position**, if Fayton hedges the currency risk related to her position in Suxzi shares using options and the GBP depreciates as expected, the effective USD value of Suxzi shares in six months would be *closest* to:

A) USD 42,000,000.

B) USD 41,310,000.

C) USD 42,150,000.

Explanation

To hedge the currency risk of Suxzi shares that are worth GBP 30,000,000, Fayton can purchase a put option at 1.4 USD/GBP. If the exchange rate depreciates as expected to 1.38 USD/GBP—a level that is lower than 1.4 USD/GBP— Fayton will exercise her right and sell the GBP for 1.4 USD. Note that the put option premium should be deducted when calculating the effective value in USD of the GBP position.

Accordingly, the effective value in USD of the Suxzi position in six months is 30,000,000 × (1.4 – 0.023) = USD 41,310,000. Thus, this is the correct answer.

The USD 42,000,000 answer option is incorrect as the USD value of the Suxzi position is calculated without accounting for the put option premium, as 30,000,000 × 1.4 = USD 42,000,000.

The USD 42,150,000 answer option is incorrect as the call option is used instead of the put option, resulting in the USD value of the Suxzi position being miscalculated as 30,000,000 × (1.42 – 0.015) = USD 42,150,000. Remember that the call option gives the right to purchase, whereas the put option gives the right to sell the underlying.

(Module 16.5, LOS 16.e)

Question #18 - 21 of 75

Question ID: 1675135

Which of the following derivatives strategies is *least appropriate* for Client A?

A) Long put. ✗

B) Zero-cost collar. ✗

C) Short straddle. ✓

Explanation

A short straddle involves selling both a call and a put option with the same strike price. It is not appropriate for Client A because it is not a hedging strategy, and it will generate significant losses if the underlying stock price declines (or increases) significantly. Therefore, the strategy performs well in stable markets.

Both a long put and a zero-cost collar (a combination of a long put, long stock, and short call) will protect against price declines, although the collar (because of the short call) will limit the strategy's upside potential.

(Module 16.6, LOS 16.f)

Question #19 - 21 of 75

Question ID: 1675136

Which of the following derivatives strategies is *most appropriate* for Client B?

A) Credit spread. ✓

B) Debit spread.

C) Protective put.

Explanation

A credit spread (bear call or bull put spread) involves selling a more expensive call (put) option and buying a less expensive call (put) option for a net inflow of premium. That satisfies Client B's preference for income in a stable market.

Neither a debit spread (bull call or bear put spread) nor a protective put would be appropriate because they both involve a net outflow of premiums, which would be inconsistent with Client B's preference for income.

(Module 16.6, LOS 16.f)

Question #20 - 21 of 75

Question ID: 1675137

Kotra decides to execute a bull put spread for Client B by selling a put for $5.00 with a strike price of $35 and buying a put for $2.20 with a strike price of $30 on one of the portfolio stocks. The breakeven price for the bull put spread is *closest* to:

A) $37.80.

B) $32.20.

C) $27.20.

Explanation

Breakeven = higher strike − net premium received = $35 − ($5.00 − $2.20) = $32.20.

(Module 16.6, LOS 16.f)

Question #21 - 21 of 75

Question ID: 1675138

In speaking with a colleague, Kotra notes that short straddles are volatility plays that have asymmetric loss potential. Kotra's statement is *most likely* correct regarding:

A) both volatility plays and asymmetric loss potential.

B) volatility plays only.

C) asymmetric loss potential only.

Explanation

Short straddles involve selling both a call and a put with the same exercise prices and maturity on the same underlying stock. They are neutrality plays (not volatility plays) because they work well in stable (flat) markets. The investor profits the most the closer the underlying stock ends up to the strike price.

The potential loss on the short put option is limited to the strike price (if the stock price falls to zero), while the potential loss on the short call option is unlimited (if the stock price rises infinitely). Therefore, the loss potential is asymmetrical.

(Module 16.6, LOS 16.f)

Question #22 of 75

Question ID: 1675186

An option trader expects that a telecommunications company will make a significant announcement tomorrow about its recent acquisition, either announcing large profits from the acquisition, or large losses. The company's shares are currently trading at $105. The news could cause a 20% change in the share price over the next week, after which the share price is expected to stabilize. Options with 1-week maturities are available. Which of the following strategies is *most appropriate* for the trader?

A) Short at-the-money straddle. ✗

B) Long at-the-money straddle. ✓

C) Bull call spread by buying a $105 strike call and selling a $126 strike call. ✗

Explanation

Given that the trader is looking to take advantage of an increase in volatility with uncertain direction, a long straddle with at-the-money calls and puts is most appropriate. This is an expensive strategy but would be beneficial if the volatility is high.

A short straddle is a neutrality play and would result in potentially large losses under highly volatile share prices. A bull call spread does not allow the trader to benefit from a decline in share price.

(Module 16.12, LOS 16.j)

Question #23 of 75

Question ID: 1675115

Suppose an investor purchased a stock for $40 last month and sold (took a short position in) a $45 call, for a $3 premium. At expiration of the call, the stock is trading at $47. Which of the following statements is *most accurate*?

A) The investor would have made a larger profit by simply owning the stock, without the option position. ✗

B) The investor has an $8 gain at expiration of the option. ✓

C) At expiration, the option is out of the money. ✗

Explanation

At expiration, the option is in the money by $47 − $45 = $2 and the call will be exercised. As a result, the investor's profit is $7 from the stock, a $3 premium from the option sale, and a $2 loss from the short call, for a total gain of $8.

The investor does not benefit from an upside from the stock above $45 because that benefit was sold to someone else through the short call option. Without the option, the investor would have had a $7 gain on the stock (= $47 − $40), which is actually less than the gain from the covered call position.

(Module 16.3, LOS 16.b)

Question #24 - 27 of 75

Question ID: 1675157

Regarding Jay's case, which of the following is *least likely* a valid reason for recommending a short position in the call option on the stock of GHY?

A) The GHY stock was undervalued and was expected to rise by $2.96 until maturity of the call option. ✓

B) The GHY stock was expected to rise to $63.30 at maturity of the call option. ✗

C) The GHY stock was expected to remain stable or decrease until maturity of the call option. ✗

Explanation

The out-of-the-money call option would have turned in the money if the stock price increased by more than 63.30 – 61.98 = $1.32, and the breakeven stock price on the short call option would have been reached if the stock price increased by 1.32 + option premium = 1.32 + 1.56 = $2.88. If the stock was determined undervalued and was expected to increase by more than $2.88, recommending to take a short position on the call would have resulted in a loss to the investor.

If the stock was expected to rise to $63.30 at maturity of the call option, the option would have expired at the money and the investor would have earned the premium ($1.56) by shorting the option.

If the stock price of GHY was expected to remain stable or decrease until maturity of the call option, the call would have expired out of the money, resulting in a gain (equivalent to the call option premium) to the party who took a short position on the call.

(Module 16.3, LOS 16.b)

Question #25 - 27 of 75

Question ID: 1675158

Regarding the case of Poet, based on **Exhibit 1: Selected Information Pertaining t o the Put Options Used in the Bear Spread**, the loss/profit on the long bear put spread is *closest* to:

A) –$7.70. ✗

B) $8.10. ✗

C) $0.40. ✓

Explanation

A long bear put spread, in which the put with the higher strike price is longed and the otherwise identical put with the lower strike price is shorted, is used to hedge against a partial decrease in the stock price. Based on **Exhibit 1: Selected Information Pertaining t o the Put Options Used in the Bear Spread**, the long bear put spread would have hedged against a decrease in the stock price until $24.50 (the put with the lower exercise price). Because the spot price is $20.71 at expiration, both the long and the short put are in the money, and the profit on the bear put spread is calculated as

$$\prod = X_H - X_L - P_L - P_H$$

where X_H is the higher strike put price; XL is the lower strike put price; p_L is the lower strike put premium; and p_H is the higher strike put premium.

$$\prod = 32.6 - 24.5 + 2.26 - 9.96 = \$0.40.$$

The –$7.70 answer option is incorrect as the loss is miscalculated as 2.26 – 9.96 = –$7.70.

The $8.10 answer option is incorrect as the profit is miscalculated by ignoring the put premiums as 32.6 – 24.5 = $8.10.

(Module 16.6, LOS 16.f)

Question #26 - 27 of 75

Question ID: 1675159

Which of the following is *most likely* the best reason for recommending Kaya to take a short calendar spread?

A) The market was expected to be stable over the near term, and its implied volatility was expected to increase over the longer term. ✗

B) The market was expected to be volatile over the near term, and its implied volatility was expected to decrease over the longer term. ✓

C) The market was expected to be volatile over the near term, and its implied volatility was expected to increase over the longer term.

Explanation

A short calendar spread involves taking a long position in a near-term put option (the June put) and a short position in a longer-term put option (the December put).

A short calendar spread is taken when the market is expected to be volatile over the near term (to benefit from the long position) and the implied volatility is expected to decrease over the longer term (to benefit from the short position), so this is the correct answer.

A long near-term put option does not benefit the investor if the market is stable over the near term, and a short longer-term put option subjects the investor to a loss if the market's implied volatility increases over the longer term.

A short calendar spread benefits the investor when the implied volatility over the longer term decreases because the longer-term put option is shorted.

(Module 16.10, LOS 16.g)

Question #27 - 27 of 75

Question ID: 1675160

Regarding Wilson's case, based on **Exhibit 2: Selected Information on KNR Call and Put Options**, the change in the value of the collar position for a $1 increase in the stock price of KNR is *closest* to:

A) $4,500.

B) $1,000.

C) −$1,900.

Explanation

A collar is a combination of a long stock, an out-of-the-money (OTM) short call, and an otherwise identical OTM long put.

Because Wilson owns 5,000 shares of KNR, the collar he took involved shorting out-of-the-money call options on 5,000 shares (the calls with the $60.20 strike price) and longing out-of-the-money put options on 5,000 shares (the puts with the $52.20 strike price).

An increase in the stock price of KNR would result in a change in the value of the following:

- Stock position by 5,000 × 1 = $5,000 (because the delta of a stock is +1).
- Short call position by –5,000 × (0.35) = −$1,750.
- Long put position by 5,000 × (−0.45) = −$2,250.

Thus, as the KNR stock price increases by $1, the value of the collar position increases by 5,000 – 1,750 – 2,250 = $1,000, so this is the correct answer.

The −$1,900 answer option is incorrect as the in-the-money (instead of the out-of-the-money) call and put options were assumed to be included in the collar, resulting in the change in the value of the collar = (5,000 × 1) – (5,000 × 0.65) + [5,000 × (–0.73)] = −$1,900.

The $4,500 answer option is incorrect as the delta of the short call position is erroneously assumed to be positive. The resulting change in the value of the collar is calculated as 5,000 + 1,750 – 2,250 = $4,500.

(Module 16.9, LOS 16.d)

Question #28 of 75

Question ID: 1675117

An investor who purchases a stock and wants the maximum price protection for her stock, but is also conscious of up-front cost, should enter into a protective put position using a(n):

A) out-of-the-money put option. ✗

B) in-the-money put option. ✗

C) at-the-money put option. ✓

Explanation

Both the at-the-money and in-the-money put options will offer full price protection because they hedge against any stock price decline below the current stock price. However, the at-the-money put option will be cheaper.

While the out-of-the-money put option is the least expensive of the three, it does not offer full downside price protection—it comes into the money only below the put strike price.

(Module 16.4, LOS 16.c)

Question #29 of 75

Question ID: 1675127

An investor purchases a stock for $38 and a put for $0.50 with a strike price of $35. The investor sells a call for $0.50 with a strike price of $40. What is the maximum profit and loss for this position?

A) Infinite profit and maximum loss = –$4.00.

B) Maximum profit = $2.00 and maximum loss = –$3.00.

C) Maximum profit = $3.00 and maximum loss = –$4.00.

Explanation

The option position described is a zero-cost collar. It is zero cost because the premium paid for the protective put is offset by the premium received for writing a covered call. The collar will put a band around the prospective returns by limiting the upside and downside of position. The upside will be limited by the strike price on the covered call ($40), while the downside will be limited by the strike price of the put ($35).

Maximum profit = $40 – $38 = $2.

Maximum loss = $35 – $38 = –$3.

(Module 16.6, LOS 16.f)

Question #30 - 33 of 75

Question ID: 1675190

Which of the following strategies will *most likely* replicate a temporary short position on CODS?

A) Short a call option and long a put option on CODS with the same exercise price and maturity.

B) Long a call option and short a put option on CODS with the same exercise price and maturity.

C) Long a call option and long a put option on CODS with the same exercise price and maturity.

Explanation

An investor who wishes to replicate a temporary short position in a stock that she owns without having to short sell the stock and then reacquire it can purchase a put option on the stock and sell a call option with the same strike price and maturity, thereby creating a *synthetic short forward position*. When the options mature, if the stock price is lower than the strike price, the investor would sell the stock by exercising the put option and realize a profit; if the stock price is higher than the strike price, the investor would have to sell the stock to the call option buyer at the strike price.

Long a call option and long a put option on CODS with the same exercise price and maturity: This represents a straddle, which does not replicate a temporary short position on an underlying asset; a straddle is implemented when the investor expects the stock price to be highly volatile, moving either upward or downward.

Long a call option and short a put option on CODS with the same exercise price and maturity: This strategy replicates a long forward position on the stock, increasing the investor's exposure to the stock instead of replicating a temporary short position on the stock.

(Module 16.1, LOS 16.a)

Question #31 - 33 of 75

Question ID: 1675191

Note 1 pertaining to the options on the VIX futures contract is *most likely*:

A) incorrect with regard to the positive profit on the strategy at expiry. ✓

B) correct. ✗

C) incorrect with regard to hedging tail risk of equity investments. ✗

Explanation

An equity investment may be hedged against an extreme decline in the market (i.e., tail risk) by buying a VIX futures contract—a futures contract that provides a proxy for implied volatility. Options are available on VIX futures contracts. At expiry, a call option will be profitable if the VIX futures price is higher than *the call strike price plus the premium paid*.

The strategy's profit is positive if the price of the VIX futures contract is higher than the strike price plus premium paid.

VIX futures contracts are often used to hedge tail risk of equity investments.

(Module 16.12, LOS 16.j)

Question #32 - 33 of 75

Question ID: 1675192

Based on Exhibit 1, the rate of return on the PERTO stock that will achieve breakeven on the strategy suggested by Wesley is *closest* to:

A) 15%.

B) 10%.

C) 17%.

Explanation

The strategy suggested by Wesley is a *bull call spread*, which is constructed by writing a call option with the higher exercise price and buying an otherwise identical call option but with a lower exercise price.

The breakeven price of a bull call strategy is calculated as

$$\text{breakeven price} = X_L + c_L - c_H$$

where X_L is lower strike price, c_L is the premium of the call option with the lower exercise price, and c_H is the premium of the call option with the higher strike price.

The breakeven price = 15 + 1.6 – 0.2 = $16.40.

As such, to achieve breakeven on the bull call strategy, the stock price should increase from $14 to $16.40, achieving a return of $\frac{16.40-14}{14}$ = 17.14%.

With the answer option of 10%, the breakeven price is miscalculated as 14 + 1.6 – 0.2 = $15.40, resulting in a return of $\frac{15.40-14}{14}$ = 10.0%.

With the answer option of 15%, the return is miscalculated as $\frac{16.40-14}{16}$ = 15.0%.

(Module 16.6, LOS 16.f)

Question #33 - 33 of 75

Question ID: 1675193

Based on Wesley's report regarding PERTO's outlook, the *most* suitable strategy is:

A) Strategy 2, which is implemented by going long a 12-month call option and going short a 3-month call option.

B) Strategy 2, which is implemented by going long a 12-month put option and going short a 3-month put option.

C) Strategy 1, which is implemented by going long a call option with an exercise price of $40 and going short a call option with an exercise price of $45.

Explanation

A *calendar spread* is implemented by buying an option and selling an otherwise identical option but with a different expiration date. Taking advantage of the time decay is a primary motivation behind calendar spreads.

A *longcalendar spread* is implemented by writing the near-term option and buying the long-term option; the premium received on the near-term option would partially offset the premium paid on the longer-term option. By using call options, because the resulting long calendar spread combines the neutral expectations over the short run (3 months) with the bullish expectations over the longer run (12 months), this is the correct answer.

While going long a 12-month put option and going short a 3-month put option is a long calendar spread, such a strategy does not consider the bullish expectations over the longer run; call options instead of put options should be used because the one -year outlook on the stock is bullish**.**

Buying an option with the lower exercise price and writing an option with the higher exercise price is referred to as a *bull spread*. When the options used are calls, the strategy is referred to as a *bull call spread*.Strategy 1 is not the most suitable because it disregards the neutral expectations over the near term (one month), and it caps the payoff at the higher exercise price if the stock price rises above it. No indication is given that the investor wishes to reduce the cost of the strategy at the expense of forfeiting potential upside gains above a particular strike price.

(Module 16.10, LOS 16.g)

Question #34 of 75

Question ID: 1675113

A portfolio manager currently oversees a $300 million equity portfolio, which is invested equally in Stock 1, Stock 2, and Stock 3. The manager expects that over the next six months, the price of Stock 2 will not deviate significantly from its current price of $60, and would like to monetize this view. There is a liquid options market for each security, and both 6-month call and put options are available. Which of the following strategies would be *most appropriate* for the manager?

A) Sell a $65 call.

B) Buy a $62 put.

C) Buy a $58 call.

Explanation

The manager can monetize Stock 2 in a flat market through a covered call strategy, which involves selling a higher strike price call option on the stock. The manager receives the premium income from the call, and benefits from any price appreciation up to $65. The downside of this strategy is that the manager would forego any price appreciation above $65.

Buying a call or a put requires paying a premium, so it would not achieve the goal of monetizing the portfolio assets. In addition, because both the $58 call and $62 put are in the money, they are expensive strategies that will lose their value quickly.

(Module 16.3, LOS 16.b)

Question #35 of 75

Question ID: 1675120

An investor has a short position in Alpha stock. The investor decides to buy a $50 strike call and sell a $50 strike put on the security. The investor is:

A) fully hedged against all price movements in Alpha. ✓

B) exposed to more upside risk than downside risk. ✗

C) exposed to more downside risk than upside risk. ✗

Explanation

Without the options, the investor with a short position in Alpha would gain from a price decline and lose when the price of Alpha increases. By purchasing a call, the investor hedges against the risk of a price increase. By selling a put, the investor transfers the potential for a gain from a price decline to the put buyer. Because the call and put are at the same strike price, the investor has effectively fully hedged against all price movements in Alpha.

(Module 16.5, LOS 16.e)

Question #36 of 75

Question ID: 1675122

A portfolio manager with a short position in an underlying security who expects no price movement in the security over the next six months is *least likely* to benefit from:

A) selling a 6-month call on the security. ✗

B) selling a 6-month put and buying a 6-month call on the security. ✓

C) selling a 6-month put on the security. ✗

Explanation

The manager can generate premium income by selling either calls or puts because the options would not be exercised in the expected flat security price market. (However, the manager is exposed to the risk that the security price could move, in which case the options would be exercised.)

The combined position of selling a put and buying a call, assuming no price movement in the underlying, is not advantageous because the income received from the put premium would be (partly) offset by the cost of buying the call option.

(Module 16.5, LOS 16.e)

Question #37 of 75

Question ID: 1675161

A *risk reversal* strategy that combines a short put with a long call creates:

A) a long exposure in the underlying shares. ✓

B) a short exposure in the underlying shares. ✗

C) a neutral exposure in the underlying shares. ✗

Explanation

Combining a short put (which obligates the seller to buy the shares if the put option is exercised) and a long call (which gives the right to buy) creates a long exposure in the underlying shares.

(Module 16.11, LOS 16.h)

Question #38 of 75

Question ID: 1675145

The shape of the protective put payoff diagram at expiration *most closely* resembles the payoff diagram of a:

A) short call. ✗

B) short stock. ✗

C) long call. ✓

Explanation

The payoff diagram at expiration of a protective put most closely resembles the payoff diagram of a long call. Both positions have limited downside risk but unlimited upside potential.

(Module 16.9, LOS 16.d)

Question #39 of 75

Question ID: 1675129

Linda Morgan is in a training program at a large investment bank. Currently, she is spending three months at her firm's derivatives trading desk. One of the traders, Jason Gover, CFA, asks her to compare different option trading strategies. Gover would like Morgan to pay particular attention to strategy costs and their potential payoffs. Morgan is not very comfortable with option models and must first investigate how to properly price European-style and American-style equity options. Gover has given her software that provides various analytical information. Morgan has decided to begin her analysis using two different scenarios to evaluate option behavior. Her scenarios are illustrated in Exhibit 1 and Exhibit 2. Note that all of the rates and yields are on a continuous compounding basis.

Exhibit 1

Exhibit 1	
Stock Price (S)	$100
Call Strike Price (X)	$100
Price	$5.51

Exhibit 2

Exhibit 2	
Stock Price (S)	$100
Put Strike Price (X)	$100
Price	$5.68

Gover instructs Morgan to consider using a straddle in which an at-the-money call and put option would be purchased. Assume all other variables remain identical.

Under what circumstance would an investor want to purchase a straddle? When the investor expects prices to:

A) increase. ✗

B) stay close to the exercise price of the options. ✗

C) increase or decrease substantially. ✓

Explanation

An investor would purchase a straddle when they expect a large movement in the price of a stock, but are unsure of the direction.

(Module 16.6, LOS 16.f)

Question #40 of 75

Question ID: 1675128

Linda Morgan is in a training program at a large investment bank. Currently, she is spending three months at her firm's derivatives trading desk. One of the traders, Jason Gover, CFA, asks her to compare different option trading strategies. Gover would like Morgan to pay particular attention to strategy costs and their potential payoffs. Morgan is not very comfortable with option models and must first investigate how to properly price European-style and American-style equity options. Gover has given her software that provides various analytical information. Morgan has decided to begin her analysis using two different scenarios to evaluate option behavior. Her scenarios are illustrated in Exhibit 1 and Exhibit 2. Note that all of the rates and yields are on a continuous compounding basis.

Exhibit 1

Exhibit 1	
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Price	$5.51

Exhibit 2

Exhibit 2	
Stock Price (S)	$100
Put Strike Price (X)	$100
Price	$5.68

Gover instructs Morgan to consider using a straddle in which an at-the-money call and put option would be purchased. Assume all other variables remain identical.

Linda now wants to compute the breakeven points for the straddle using the options and underlying stock in Exhibits 1 and 2. Which of the following are the *closest* to the breakeven points for the straddle?

A) $95.58, $104.42.

B) $93.11, $106.89.

C) $88.81, $111.19.

Explanation

This is the exercise price plus/minus the maximum loss. Because the total cost of the straddle is $11.19, the breakeven points are $100 +/– 11.19.

(Module 16.6, LOS 16.f)

Question #41 of 75

Question ID: 1675147

A long position in an asset, combined with a short forward on the same asset (in the same amount), has a delta of:

A) 0 to +1.

B) 0.

C) −1 to 0.

Explanation

The delta of a long asset is +1, whereas the delta of a short forward on the same asset is −1. The delta of the overall position is, therefore, 0 (full hedge).

(Module 16.9, LOS 16.d)

Question #42 - 44 of 75

Question ID: 1675180

Here is the following scenario:

- Performance to date: up 3%
- Client objective: to maintain a positive stock position and retain upside potential
- Austin's scenario: expect low stock price volatility between now and end of year

Which is the *best* option strategy to meet the client's objective?

A) Protective put.

B) Bull call.

C) Covered call.

Explanation

The client wants to stay positive on the stock, and a protective put will retain the stock upside with limited downside risk. In addition, volatility is low, which will make option prices low. Both of the other strategies will compromise stock upside potential and involve selling options to reduce initial investment cost. Lowering the initial investment was not a specific goal, and it makes little sense to do so while option prices are low.

(Module 16.12, LOS 16.i)

Question #43 - 44 of 75

Question ID: 1675181

Here is the following scenario:

- Performance to date: up 16%
- Client objective: earn at least 15%
- Austin's scenario: good chance of large gains or large losses between now and end of year

Which is the *best* option strategy to meet the client's objective?

A) Long straddle. ✓

B) Collar. ✗

C) Short straddle. ✗

Explanation

A long straddle produces gains if prices move up or down, and limited losses if prices do not move. A short straddle produces significant losses if prices move significantly up or down. A collar (combination of a protective put and covered call) is a bet on low volatility, with limited downside and limited upside potential. Therefore, the collar does not produce large gains or losses.

(Module 16.12, LOS 16.i)

Question #44 - 44 of 75

Question ID: 1675182

Here is the following scenario:

- Performance to date: up 16%
- Client objective: earn at least 15%
- Austin's scenario: good chance of large losses between now and end of year

Which is the *best* option strategy to meet the client's objective?

A) Long call options.

B) Long put options.

C) Short call options.

Explanation

Long put positions gain when stock prices fall and produce very limited losses if prices instead rise. Short calls also gain when stock prices fall, but they create losses if prices instead rise. The other two positions will not protect the portfolio, should prices fall.

(Module 16.12, LOS 16.i)

Question #45 of 75

Question ID: 1675119

An investor purchases Zalaa stock at $74.40, and at the same time buys a $73 put option on Zalaa for $2.60. Which of the following statements is *most accurate*?

A) The breakeven is $75.60.

B) The maximum loss is $4.00.

C) The maximum gain is $2.60.

Explanation

The investor's maximum loss is $4.00, which is $1.40 loss on the stock (= $73 − $74.40) plus the $2.60 put premium paid.

The breakeven is at $77.00 (= $74.40 + $2.60). The maximum gain on a protective put is unlimited because the investor will simply let the option expire if the stock price rises, and the potential increase in the stock price is infinite.

(Module 16.4, LOS 16.c)

Question #46 of 75

Question ID: 1675183

An investor is interested in buying KPQ shares but finds the current trading price of $128 too high. The investor is willing to pay $115 per share. Which of the following strategies is *most appropriate* for the investor?

A) Buy $115 strike price calls.

B) Sell $115 strike price puts.

C) Enter into a $115 strike price collar.

Explanation

The investor would receive premiums from selling the $115 strike price puts. If the share price declines below $115, the investor would be obligated to buy the shares at $115, although this price would be reduced by the option premium received from selling the puts. The investor would keep all future upside potential once the shares are purchased.

The $115 strike price calls would be in the money by $13 each. This is equivalent to buying the shares at $128 (in fact, it is more expensive given that the options would also have time value) and is, therefore, not appropriate. A collar would involve buying the shares at $128, and it would also limit the investor's upside potential and is not appropriate.

(Module 16.12, LOS 16.i)

Question #47 of 75

Question ID: 1675124

Assume that the current price of a stock is $100. A call option on that stock with an exercise price of $97 costs $7. A call option on the stock with the same expiration and an exercise price of $103 costs $3. Using these options, what is the profit for a long bull spread if the stock price at expiration of the options is equal to $110?

A) $2. ✓

B) –$2. ✗

C) $6. ✗

Explanation

The buyer of a bull spread buys the call with an exercise price below the current stock price and sells the call option with an exercise price above the stock price. Therefore, for a stock price of $110 at expiration of the options, he gets a payoff $13 from his long position and a payoff of –$7 from his short position for a net payoff of $6. The cost of the strategy is $4. Hence, the profit is equal to $2.

(Module 16.6, LOS 16.f)

Question #48 of 75

Question ID: 1675139

Melinda has a long position on 50,000 of DWD, Inc. (DWD). DWD is currently trading at $42 per share, and it is widely believed that DWD's price will not rise in the next year. She is concerned that the price could fall below $35. Assuming Melinda is seeking protection of her long position for the next year and wants to minimize her total costs, what is her *least likely* course of action?

A) Purchase a put with a strike price of $41 and sell a put with a strike price of $35.

B) Purchase a put with a strike price of $40.

C) Purchase a put with a strike price of $39 and sell a call with a strike price of $45.

Explanation

Purchasing a put with a strike price of $41 will provide her with downside protection if the stock price falls from the current price of $42. However, selling a put with a strike price of $35 exposes her to losses if the stock price falls below $35, just like she is concerned about. Therefore, purchasing a 41 put and selling a 35 put is not an appropriate course of action.

Purchasing a put with a strike price of $39 will provide her with downside protection if the stock price falls from the current price of $42. Selling a call with a strike price of $45 will provide her with some additional income because the stock price is not expected to rise. Therefore, this is a reasonable course of action.

Purchasing a put with a strike price of $40 will provide her with downside protection if the stock price falls from the current price of $42. Therefore, this is a reasonable course of action.

(Module 16.6, LOS 16.f)

Question #49 of 75

Question ID: 1675116

Joe purchases a stock for $55 and simultaneously purchases a put for $2 that has a strike price of $45. Which of the following statements is correct?

A) The breakeven price is $47.

B) The maximum loss is $8.

C) The maximum profit is unlimited.

Explanation

The maximum profit is the future stock price less the cost of the stock ($55) less the put premium ($2). Theoretically, the future stock price is unlimited, and so the maximum profit is also unlimited.

The maximum loss is the difference between the cost of the stock and the strike price ($55 – $45) plus the put premium ($2), for a total of $12.

The breakeven price is the cost of the stock ($55) plus the put premium ($2), for a total of $57.

(Module 16.4, LOS 16.c)

Question #50 of 75

Question ID: 1675131

A portfolio manager of a large-cap equity fund is convinced that one of her portfolio holdings will exhibit extremely low price volatility over the next three months. The manager has a very high risk tolerance. Which of the following options strategies would provide the maximum potential profit from her forecast?

A) Collar with a higher call premium than put premium. ✗

B) Short straddle. ✓

C) Credit spread with calls. ✗

Explanation

All three strategies are credit strategies, meaning that they result in a net cash inflow to the manager. However, the short straddle will provide the maximum gain because it involves selling both a call and a put, benefiting from receiving two option premiums. Assuming the manager's forecast of no or very low volatility is correct, the options would remain out of the money at expiration and would not be exercised. Both the spread and collar involve buying and selling options simultaneously, and they would not result in the largest cash inflow to the manager.

Note that the short straddle also has the largest potential for loss, and is therefore the riskiest strategy—although this is not a constraint in the question, given the manager's high risk tolerance.

(Module 16.6, LOS 16.f)

Question #51 of 75

Question ID: 1675188

An investor is expecting low volatility in the near term for the price of a stock she owns but has a long-term bullish view. The *best* course of action for the investor to take would be to enter into a:

A) long calendar spread with calls. ✓

B) long zero-cost collar. ✗

C) short straddle. ✗

Explanation

A long calendar spread involves buying a longer-dated call and selling a shorter-dated call with the same strike and underlying. This allows a longer-term bullish view that is subsidized by the premium of the shorter-dated call.

A collar caps the upside and is therefore not appropriate. A short straddle will result in premium income but is very risky if the share price moves significantly in either direction.

(Module 16.12, LOS 16.j)

Question #52 of 75

Question ID: 1675149

In May, BCA shares are trading at $43.50. An investor enters into a long calendar spread on BCA by buying an October 45 call for $5.50 and selling a July 45 call for $2.50. In July, the short call expires out of the money. At the time of the short call expiry, the investor would have an overall gain on the spread only if:

A) the BCA stock price increased above $45. ✗

B) the BCA stock price did not fall below $40.50. ✗

C) the October call is worth more than $3. ✓

Explanation

In May, the investor paid a net cost of $5.50 − $2.50 = $3.00 for the calendar spread (premium income from the short call subsidized a portion of the long call premium). In July, the investor would have a profit if the October long call value is at least $3.00, which is the investor's initial net cost.

Because this is a calendar spread, there is no ownership of the BCA shares; the investor did not buy the BCA shares for $43.50. As a result, the share price does not need to increase above $45 or avoid falling below $40.50 for the investor to generate a profit.

(Module 16.10, LOS 16.g)

Question #53 of 75

Question ID: 1675110

An investor who is long a put and short a call on the same underlying (with the same strike price and same expiration) has an identical payoff position to someone who is:

A) long the underlying stock and lends the present value of the option strike price. ✗

B) short the underlying stock and lends the present value of the option strike price. ✓

C) long the underlying stock and borrows the present value of the option strike price. ✗

Explanation

The put-call parity relationship describes two positions that have identical payoff values. Under the put-call parity, a long call and a short put (on the same underlying stock, and with the same exercise price and expiry) have an identical payoff to taking a long position in the stock and borrowing the present value (PV) of the strike price:

call premium − put premium = underlying stock − PV(X) received

We can rearrange this formula to reflect the situation in this question:

put premium − call premium = PV(X) received − underlying stock

In other words, a long put and short call position can be replicated through lending the PV of the strike price plus a short position in the underlying stock.

(Module 16.1, LOS 16.a)

Question #54 of 75

Question ID: 1675118

An option trader establishes a protective put position by buying a stock at $85 and buying an $86 put option for $2. The breakeven stock price for this position is:

A) $3 above the initial stock price.

B) $1 below the initial stock price.

C) $2 above the initial stock price.

Explanation

The breakeven stock price is $87, which is $2 (= the option premium) above the initial stock price. At $87, the trader has a $2 gain on the stock, offset by the $2 premium paid for the put option. (Note that at $87, the put option is still out of the money.)

(Module 16.4, LOS 16.c)

Question #55 of 75

Question ID: 1675111

Richard Macpherson recently entered into a six-month forward contract to buy 400 shares of Mobilia shares at $85 per share. Mobilia does not pay dividends. Which of the following strategies would be *most effective* as a synthetic hedge of Macpherson's position?

A) Buy a six-month $85 call and sell a six-month $85 put on Mobilia.

B) Buy a six-month $85 call and buy a six-month $85 put on Mobilia.

C) Sell a six-month $85 call and buy a six-month $85 put on Mobilia.

Explanation

A long forward contract can be hedged using a synthetic short forward contract. A synthetic short forward can be created by buying a put option and selling a call option with the same maturity and price as the forward contract.

(Module 16.1, LOS 16.a)

Question #56 of 75

Question ID: 1675148

A long calendar spread is similar to a bull spread in that both spreads:

A) involve taking a position in the underlying shares. ✗

B) involve buying and selling both calls and puts. ✗

C) are beneficial if there is relatively little price movement in the underlying shares. ✓

Explanation

Both a long calendar spread and a bull (call or put) spread is beneficial if there is relatively small movement in the underlying shares. A calendar spread may involve buying a longer-dated call and selling a shorter-dated call, both of which are close to being at the money (ATM).

Spreads involve buying and selling either calls only or puts only. They do not require taking a position in the underlying shares.

(Module 16.10, LOS 16.g)

Question #57 - 60 of 75

Question ID: 1675170

Regarding Question 1, the delta of the covered call will *most likely* be:

A) lower than 0 at initiation and close to –1 at expiry of the call option. ✗

B) close to 0 at initiation and close to 0 at expiry of the call option. ✗

C) higher than 0 at initiation and close to 0 at expiry of the call option. ✓

Explanation

A covered call is constructed by going long a stock and short a call option on the stock. At initiation, because the call option is out of the money, the option's delta will be lower than 1 (typically less than 0.5). Thus, because the stock's delta is 1 and only partially hedged by the negative delta of the short out-of-the-money call option, the delta of the covered call strategy is positive at initiation.

At expiry, because the call is in the money, the delta of the short call will be close to -1. The delta of the short call option in this case will offset the delta of the stock, and the resulting delta of the covered call strategy will be closer to 0.

(Module 16.9, LOS 16.d)

Question #58 - 60 of 75

Question ID: 1675171

Which of the following is the *most accurate* answer to Question 2?

A) The market sentiment is bullish, and a suitable strategy may involve buying call options. ✗

B) The market sentiment is bearish, and a suitable strategy may involve writing call options. ✗

C) The market sentiment is bearish, and a suitable strategy may involve buying put options. ✓

Explanation

An increase in the level of volatility skew (i.e., the difference in the implied volatility between the out-of-the-money put options and the out-of-the-money call options) associated with an increase in absolute implied volatility indicates that the market sentiment is bearish. Accordingly, buying a put option may be a suitable strategy.

An increase in both the level of volatility skew and absolute implied volatility indicates the market is bearish.

Because options are positively related to implied volatility, selling options leads to losses if implied volatility increases.

(Module 16.11, LOS 16.h)

Question #59 - 60 of 75

Question ID: 1675172

Based on **Exhibit 1: Selected Information on 3-Month Stabru* Options (In EUR)** and the report of OPM's equity research team on Stabru, the maximum profit on a suitable collar strategy for Zena is *closest* to:

A) EUR 9.8.

B) EUR 4.8.

C) EUR 5.9.

Explanation

The combination of a stock, a long put option with an exercise price below the current stock price, and a short call option with an exercise price above the current stock price is referred to as a *collar*.

Because Zena wishes to hedge against a decrease in the stock price below EUR 65, a put option with a strike price of EUR 65 should be bought. Because the stock price of Stabru is not expected to increase above EUR 75 within three months, selling a 3-month call option with an exercise price of EUR 75 may be considered appropriate for offsetting (or partially offsetting) the put premium paid.

The maximum profit on a collar strategy is calculated as

$$\text{maximum profit} = X_H - S_0 - p_0 + c_0$$

where X_H is the higher strike price (i.e., the strike price of the call option), S_0 is the stock price at Time 0, p_0 is the premium of the put option, and c_0 is the premium of the call option.

Maximum profit = 75 – 70 – 0.4 + 0.2 = EUR 4.8. Thus, this is the correct answer.

With the EUR 5.9 answer option, the EUR 70 call option instead of the EUR 75 call option and the exercise price of the put instead of the current stock price are used, so the result is the maximum profit = 70 – 65 – 0.4 + 1.3 = EUR 5.9.

With the EUR 9.8 answer option, the maximum profit is calculated using the exercise price of the put instead of the current stock price as 75 – 65 – 0.4 + 0.2 = EUR 9.8.

(Module 16.6, LOS 16.f)

Question #60 - 60 of 75

Question ID: 1675173

Which of the following *best* replicates a protective put on 40,000 Stabru shares for small changes in Stabru's stock price?

A) Strategy 3.

B) Strategy 1.

C) Strategy 2.

Explanation

Because small changes in the price of the underlying stock are expected, the strategy that best replicates the protective put is the strategy that has a delta similar to that of the protective put.

First, calculate the delta of the protective put as

delta of the protective put = delta of stock + delta of the at-the-money put option

Delta of the stock = 1.

Delta of the at-the-money put option ≈ –0.5.

Delta of the protective put ≈ 1 – 0.5 ≈ 0.5 per each share.

Total delta of the protective put = 40,000 × 0.5 ≈ 20,000.

The delta of a forward contract on the stock of Stabru is equal to 1 because the stock is a non-dividend-paying stock.

The delta of Strategy 1 = delta of 40,000 stocks + delta of the short forward on 20,000 stocks = 40,000 –20,000 = 20,000.

Because Strategy 1 has a delta that is approximately equal to the delta of the protective put, the change in value of Strategy 1 is similar to the change in value of a protective put for small changes in the stock price. Thus, for small changes in the stock price of Stabru, the combination of longing 40,000 Stabru shares and shorting a forward contract on 20,000 Stabru shares closely replicates a protective put strategy on 40,000 Stabru shares that uses at-the-money put options.

Because the delta of Strategy 2 is equal to 40,000 – 40,000 = 0, it does not match the delta of the protective put using at-the-money put options. As such, the changes in value of Strategy 2 and the protective put will differ for small changes in the price of the underlying stock. Therefore, Strategy 2 does not closely replicate a protective put strategy using at-the-money put options on 40,000 Stabru shares.

With the Strategy 3 answer option, a n out-of-the-money call option has a delta that is typically lower than |–0.5|. As such, the delta of Strategy 3 is higher than 40.000 – (0.5 × 20,000) = 30,000 and differs from the delta of a protective put. Therefore, Strategy 3 does not closely replicate a protective put strategy using at-the-money put options on 40,000 Stabru shares.

(Module 16.9, LOS 16.d)

Question #61 of 75

Question ID: 1675125

Assume that the current price of a stock is $100. A call option on that stock with an exercise price of $97 costs $7. A call option on the stock with the same expiration and an exercise price of $103 costs $3. Using these options, what is the expiration profit of a bear call spread if the stock price is equal to $110?

A) –$2. 

B) $2. 

C) –$6.

Explanation

The trader of a bear call spread sells the call with an exercise price below the current stock price and buys the call option with an exercise price above the stock price. Therefore, for a stock price of $110 at expiration of the options, the buyer realizes a payoff of –$13 from his short position and a positive payoff of $7 from his long position for a net payoff of –$6. The revenue of the strategy is $4. Hence, the profit is equal to –$2.

(Module 16.6, LOS 16.f)

Question #62 of 75

Question ID: 1675130

An investor has constructed a zero-cost collar on a share with a price P_0. For any movement in the share price above P_0, the investor will *most likely* have:

A) either a gain or a loss, depending on the call strike price.

B) a gain.

C) a loss until the call strike price is reached.

Explanation

A zero-cost collar is constructed on a share by selling a (typically out-of-the-money, or OTM) call and buying a put (typically also OTM) with the same premium as the call. Because the option premiums are the same, there is zero net premium, and any increase in the share price above the initial price of P_0 will result in a gain, with the gain capped at the call strike price. (Similarly, any decline in value below P_0 will result in a loss, with the loss capped at the put strike price.)

(Module 16.6, LOS 16.f)

Question #63 of 75

Question ID: 1675126

What is the expiration payoff of a long straddle, with an exercise price of $100, if the underlying stock price is $125?

A) $25.

B) $0.

C) –$25.

Explanation

A long straddle consists of a long call and put with the same exercise price and the same expiration; at a stock price of $125, the put will expire worthless, and the call value will be $25.

(Module 16.6, LOS 16.f)

Question #64 - 67 of 75

Question ID: 1675175

Regarding Wilson's notes, based on **Exhibit 1: Selected Information Pertaining to the Call Option on FNR***, which of the following is *most accurate*?

A) Note 1 is correct.

B) Note 2 is correct.

C) Both notes are correct.

Explanation

Note 1 is correct. If Tailor shorts the call option, he would receive the call premium of $7 and would be obliged to sell his 100 FNR shares if the stock price rises above the exercise price of $200. As such, Tailor would realize a profit if the stock price at expiry of the call option is below 200 + 7 = $207; if the stock price is expected to rise above $207 at the expiry of the call option, Taylor is better off not writing the call option.

Note 2 is incorrect. The breakeven price—the price prevailing in one year that would result in Tailor being indifferent between selling his shares now and writing the call option—is calculated as $S_0 - c_0 = 203 - 7 = \$196$. As such, Tailor would be better off selling his FNR shares right now at $203 rather than writing the call option if the stock price is expected to decrease by more than the premium ($7), declining below $196.

(Module 16.3, LOS 16.b)

Question #65 - 67 of 75

Question ID: 1675176

Which of the following is *most likely* the cheapest strategy that will fully protect Tailor against a decline in the price of biotech below $45?

A) Startegy 3.

B) Startegy 2.

C) Startegy 1.

Explanation

A *collar* is implemented by going long the stock, buying a put option with an exercise price below the current stock price, and writing a call option with an exercise price above the current stock price.

The collar is implemented by buying the put option with a $45 exercise price (to fully protect against a decline in the stock price below $45) and writing the call option with a $55 exercise price (to earn a premium income that would minimize the cost of the protection). Writing the call option will cap upside gains at a stock price of $55. However, because it is not mentioned that participating in upside gains is a priority (rather, implementing the cheapest possible strategy is the top priority), Strategy 2 is the recommended strategy and is the correct answer.

Strategy 1 is a protective put strategy. While both Strategy 1 and Strategy 2 provide full protection against a decline in the stock prices below $45, Strategy 2 is the lower- cost strategy.

With Strategy 3, the combination of a long put option and a short put option that has a lower exercise price ($45) is referred to as a *long bear spread*, which does not provide protection for a decline in the stock price below $45.

(Module 16.6, LOS 16.f)

Question #66 - 67 of 75

Question ID: 1675177

Based on **Exhibit 2: Selected Information Pertaining to the Options in the Straddle on MOTTO**, the breakeven prices for the short straddle strategy on the MOTTO stock are *closest* to:

A) $94.20 and $105.80. ✓

B) $97.40 and $103.20. ✗

C) $96.80 and $102.60. ✗

Explanation

A short straddle, which is implemented by writing both a put and a call with the same underlying characteristics, is profitable when the price of the underlying does not significantly change in either direction.

The breakeven prices for a straddle are calculated as

$$\text{Breakeven price}_{low} = X - c_0 - p_0$$

$$\text{Breakeven price}_{high} = X + c_0 + p_0$$

$$\text{Breakeven price}_{low} = 100 - 3.2 - 2.6 = \$94.20.$$

$$\text{Breakeven price}_{high} = 100 + 3.2 + 2.6 = \$105.80.$$

With the $96.80 and $102.60 answer option, the breakeven prices are incorrectly calculated as breakeven $\text{price}_{low} = 100 - 3.2 = \$96.80$, and breakeven $\text{price}_{high} = 100 + 2.6 = \$102.60$.

With the $97.40 and $103.20 answer option, the breakeven prices are miscalculated as breakeven $\text{price}_{low} = 100 - 2.6 = \$97.40$, and breakeven $\text{price}_{high} = 100 + 3.2 = \$103.20$.

(Module 16.6, LOS 16.f)

Question #67 - 67 of 75

Question ID: 1675178

Wilson's statement recommending a short position in a risk reversal based on his expectation that the S&P 500's level of volatility skew is expected to decline is *most likely*:

A) incorrect regarding a short position in a risk reversal being profitable if the level of volatility skew declines. ✓

B) incorrect regarding an expected decline in the level of volatility skew implying that the market is expected to turn bullish. ✗

C) correct. ✗

Explanation

Volatility skew is a common shape of the implied volatility curve where the implied volatility of out-of-the-money (OTM) put options is higher than that of OTM call options.

A decline in the level of volatility skew results from a decrease in the implied volatility of OTM put options and/or an increase in the implied volatility of OTM call options. Expecting a decline in the level of volatility skew, an investor may choose to long an OTM, undervalued call option and short an OTM, overvalued put option—a strategy that is referred to as *long risk reversal*.

A long position in a risk reversal is suitable when the level of volatility skew is expected to decline.

An expected sharp decline in the level of volatility skew indicates that the market is expected to turn bullish because the demand for OTM call options relative to the demand for OTM put options is expected to increase as investors are increasingly interested in taking on upside exposure.

(Module 16.11, LOS 16.h)

Question #68 of 75

Question ID: 1675162

Consider a risk reversal strategy with a delta of 0.45 involving 5 call contracts and 5 put contracts. Each contract is worth 100 shares. Which of the following trades would *best* ensure a delta hedged position?

A) Sell 225 shares. ✓

B) Sell 450 shares. ✗

C) Buy 450 shares. ✗

Explanation

A delta of 0.45 implies that for each $1 fall in the share price, the risk reversal would lose $1 × 0.45 × 500 shares = $225. Because the risk reversal delta is positive, the risk reversal strategy has a long exposure to the underlying shares. A delta hedge involving selling 225 shares (= 0.45 × 500) would remove the net exposure to the stock.

(Module 16.11, LOS 16.h)

Question #69 of 75

Question ID: 1675150

Which of the following statements about short calendar spreads is *most accurate*?

A) A short calendar spread is a credit spread, not a debit spread. ✓

B) A short calendar spread benefits from an increase in implied volatility. ✗

C) A trader benefits most if the shorter-dated option expires at the money (ATM).

Explanation

A short calendar spread is a credit spread, meaning that a trader would receive a net income from entering into a position because the premium paid on the long shorter-dated option is less than the premium received from selling the longer-dated option.

A short calendar spread benefits from a *decrease* in implied volatility because it would make the short position more valuable. If the shorter-dated option expires ATM, the longer-dated premium would rise, which is a *loss* on the short position.

(Module 16.10, LOS 16.g)

Question #70 - 73 of 75

Question ID: 1675165

The breakeven stock price for the option in Strategy A is *closest* to:

A) 15.21.

B) 169.21.

C) 154.00.

Explanation

To capture stock price appreciation, Harris will buy the December 154 call at a cost of 15.21 per share. The instructions are to ignore the stock holding. This is a simple, naked long call. The total initial investment is 15.21. The max loss occurs when the stock is at the strike price of 154. The option is worthless, and the 15.21 premium is lost. From 154, the stock must increase to reach breakeven. It must increase to 154 + 15.21 = 169.21.

(Module 16.1, LOS 16.a)

Question #71 - 73 of 75

Question ID: 1675166

The max gain for the bear spread using call options in Strategy B is *closest* to:

A) 3.95.

B) 12.95.

C) 4.05.

Explanation

To construct the September bear spread using calls, Harris will sell the September 150 and buy the September 158 call. He will receive 12.95 and pay 9.00 for a net receipt of 3.95. His max gain occurs if the stock declines to 150. Both options expire worthless, and he keeps the 3.95 as his max gain.

(Module 16.6, LOS 16.f)

Question #72 - 73 of 75

Question ID: 1675167

The initial shares of BIM required to hedge the 10,000 short September puts with a strike price of 152 is *closest* to:

A) short 4,080 shares. ✗

B) long 4,670 shares. ✗

C) short 4,670 shares. ✓

Explanation

The dealer will lose on the short put position if BIM declines. A short position in the stock will produce offsetting gains. The number of shares is delta × number of options. The September 152 put delta is 0.467; therefore, 10,000 × 0.467 = 4,670 shares are required. The sign on delta is irrelevant to our analysis, as option fundamentals dictate we must be short shares because a short put produces losses, as the underlying declines and a short position in the stock would produce offsetting gains.

(Module 16.11, LOS 16.h)

Question #73 - 73 of 75

Question ID: 1675168

Rebalancing the hedge on the 10,000 short September puts with a strike price of 152 is *most likely* to produce an immediate net profit if:

A) the BIM stock price decreases and expected volatility increases. ✗

B) the BIM stock price increases and expected volatility decreases. ✓

C) the BIM stock price increases and expected volatility increases. ✗

Explanation

Delta hedging a short put position requires continually rebalancing the number of shares for the hedge. As the stock price changes, delta will change and shares will have to be bought or sold to maintain the hedge. One of the practical problems of delta hedging a short option position (be it puts or calls) is that you will have to buy shares if they have gone up and sell shares if they have gone down. There is an immediate loss on the rebalancing transaction in either direction.

A decline in volatility is the only way it is possible there could be a gain. A decline in volatility lowers the value of both call and put options. That drop in value (price) is a gain to the short put (or short call) position.

(Module 16.11, LOS 16.h)

Question #74 of 75

Question ID: 1675133

An investor is short 10 contracts of July $40 calls and short 10 contracts of July $40 puts on Alphastar shares. The investor does not own the underlying shares. Which of the following statements about the investor's position is *most accurate*?

A) The investor's risk exposure to a fall in Alphastar's share price is limited rather than unlimited. ✓

B) The investor is fully hedged against changes in Alphastar's share price. ✗

C) Increased volatility of Alphastar shares will result in rising call option values but falling put option values. ✗

Explanation

The investor has limited risk, should the share price fall: the maximum loss is the $40 exercise price. However, the investor is exposed to unlimited risk if the share price were to rise because the price increase is not capped. (Note that this short position in an equal number of calls and puts on the same underlying with the same expiry is called a short straddle position.)

The investor is not hedged against price changes in Alphastar shares. In fact, the investor will suffer losses if the share price declined or increased by more than the premiums received for the options. A rise in volatility will result in both rising call and put option prices (negative to the investor).

(Module 16.6, LOS 16.f)

Question #75 of 75

Question ID: 1675146

To fully hedge against price movements in a stock, an investor should enter into a short forward position where the number of short forward shares is:

A) less than the number of shares owned.

B) greater than the number of shares owned.

C) equal to the number of shares owned.

Explanation

The delta of a long stock is +1, whereas the delta of a short forward on the same stock is −1. If the number of short forwards shares equals the number of long shares owned, the two deltas offset each other, fully protecting the investor against movements in the stock price.

(Module 16.9, LOS 16.d)